

Harshavardhana Gowda

Department of Electrical and Computer Engineering
University of California, Davis

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 Website

EDUCATION

- Ph.D. **University of California, Davis**
Electrical and Computer Engineering
Expected: Winter 2027
- B.S. **Indian Institute of Space Science and Technology**
Avionics, Fall 2018

RESEARCH EXPERIENCE

- 2023– **University of California, Davis**
Speech Neuroengineering and Cybernetics Laboratory
Graduate Student Researcher
- 2018–2022 **Indian Space Research Organization**
Satellite Communication and Navigation Laboratory
Research Scientist

RESEARCH AREAS

Speech neuromotor interfaces, wearable bio-interfaces.
Spoken language understanding, speech synthesis and recognition.
Human-computer interaction, understanding and creating embodied intelligence.

PUBLICATIONS

Peer Reviewed Articles

- 2026 **Harshavardhana T. Gowda**, Daniel C. Comstock, Lee M. Miller.
emg2speech: synthesizing speech from electromyography using self-supervised speech models.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers).  Project Page.
- 2026 **Harshavardhana T. Gowda**, Lee M. Miller.
Non-invasive electromyographic speech neuroprosthesis: a geometric perspective.
Findings of the Association for Computational Linguistics: ACL 2026.  Project Page.
- 2025 **Harshavardhana T. Gowda**, Zachary D. McNaughton, Lee M. Miller.
Geometry of orofacial neuromuscular signals: speech articulation decoding using surface electromyography.
Journal of Neural Engineering.  Project Page.

- 2025 **Harshavardhana T. Gowda**, Neha Kaul, Carlos Carrasco, Marcus Battraw, Safa Amer, Saniya Kotwal, Selena Lam, Zachary D. McNaughton, Ferdous Rahimi, Sana Shehabi, Jonathon Schofield, Lee M. Miller.
A database of upper limb surface electromyogram signals from demographically diverse individuals.
Scientific Data.  [Project Page](#).
- 2024 **Harshavardhana T. Gowda**, Lee M. Miller.
Topology of surface electromyogram signals: hand gesture decoding on Riemannian manifolds.
Journal of Neural Engineering.  [Project Page](#).

TALKS

- 2026 *Non-invasive speech neuroprostheses*
University of California, Berkeley.
25th Anniversary of the CITRIS and the Banatao Institute. Theme: “*Engineering a future fit for Humanity*”.

GRANTS AND AWARDS

Competitive Grants and Fellowships

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| 2026 | <i>Summer Graduate Research Fellowship</i> at the University of California, Davis | \$22,485 |
| 2025 | <i>NeuralStorm Fellowship</i> at the University of California, Davis | \$66,551 |
| 2024 | <i>Department of Space Fellowship</i> at the Indian Institute of Space Science and Technology | \$10,000 |

TEACHING

University of California, Davis

- Fall 2024 Electronic Circuits and Systems. Graduate Student Instructor for 100+ students. Taught implementation of electronic circuits.

SERVICE

Academic Journal Peer Review

- Journal of Neural Engineering × 3
Scientific Reports × 2
Scientific Data × 1
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026) × 4